Creating Confidence

Specs increase confidence in your project. We're not talking about the simplistic view that if your specs pass, your program is bug-free (we've been programming in the real world for far too long to believe that fairy tale).

When we say *confidence*, we mean that a well-written spec can provide evidence in favor of certain claims about your code. For instance:

- The "happy path" through a particular bit of code behaves the way you want it to.
- A method detects and reacts to an error condition you're anticipating.
- That last feature you added didn't break the existing ones.
- You're making measurable progress through the project.

None of these statements are absolutes. You're not going to prove them true by writing enough specs. But a well-crafted suite can give you enough confidence to know when you can move on to the next piece you're working on.

Eliminating Fear

Think back to a past project you've worked on that was downright *scary* to work on. What made it so frightening?

Here are some of the things we've encountered:

- A simple, innocuous change would break distant parts of the code that seemed unrelated.
- Developers felt paralyzed by fear, unable to safely make any changes to the code at all.

Working on a project with a good set of specs is a refreshing change from this situation. With broad test coverage, developers find out early if new code is breaking existing features. When your specs have your back, you can fearlessly refactor tricky bits of code, ensuring your code base does not stagnate. Speaking of refactoring....

Enabling Refactoring

Truth is rarely permanent in software projects. Your understanding of the problem domain will improve as you discover new facts. Your startup may pivot. Old assumptions embedded into the code may make implementing new features difficult.